Cavity-induced persistent current in a π loop

Model. We consider a superconducting (SC) loop of L sites closed by an interacting quantum dot (site 0) and threaded by the magnetic flux of a single lumped LC mode [Fig. 1(a)]. The full microscopic Hamiltonian is

$$\begin{aligned} H = {} & \hbar\omega\, a^\dagger a - t \sum_{r=1}^{L-2} \sum_{\sigma} \left(c^\dagger_{r+1\sigma} c_{r\sigma} + \text{h.c.} \right) \\ & + \Delta \sum_{r=1}^{L-1} \left(c^\dagger_{r\uparrow} c^\dagger_{r\downarrow} + \text{h.c.} \right) + \epsilon_d\, n_0 + U n_{0\uparrow} n_{0\downarrow} \\ & - t_J \sum_{\sigma} \left(c^\dagger_{0\sigma} c_{L-1\,\sigma} + \text{h.c.} \right) \\ & - t_J \sum_{\sigma} \left(e^{i\varphi_{\rm cl}}\, e^{ig(a+a^\dagger)}\, c^\dagger_{1\sigma} c_{0\sigma} + \text{h.c.} \right). \end{aligned} \tag{1}$$

The first two lines are the resonator, an s-wave BCS chain (hopping t, pairing Δ) and the Anderson impurity (level $\epsilon_d = -U/2$, on-site repulsion U) tunnel-coupled (t_J) to the two ends of the chain; the last line carries the total Peierls phase of the loop, which by a gauge choice is placed on a single bond. The vector potential of the mode enters through the gauge-invariant displacement operator $D = e^{ig(a+a^\dagger)}$, so diamagnetic (A^2) terms are included to all orders; $g = 2\pi\Phi_{\rm zpf}/(h/e)$ is the electron phase generated by the zero-point flux of the mode, and $\varphi_{\rm cl}$ is a static external flux. An electron phase $\varphi = \pi$ corresponds to one superconducting flux quantum $\Phi_0 = h/2e$. For a lumped resonator of impedance Z, galvanically sharing its inductance with the loop, $(2g)^2 = \pi Z/R_Q$ with $R_Q = h/4e^2$. Unless stated otherwise we take $\Delta = t$ and quote energies in units of Δ (currents in units of $e\Delta/\hbar$), $t_J = 0.5\Delta$ (level width $\Gamma = 2t_J^2/t = \Delta/2$), $L = 10$, $\hbar\omega = 7.39\times10^{-4}\Delta$ and *no* external flux, $\varphi_{\rm cl} \to 0^+$ (a seed $\varphi_{\rm cl} = 10^{-3}$ is used to select one of the degenerate states). The ring is longer than the coherence length $\xi \simeq t/\Delta$: the π-sector Josephson energy of the $U = 2\Delta$ junction, $E_2 = 7.39\times10^{-3}\Delta$, changes by 0.2% between $L = 8$ and $L = 14$, and its single-electron harmonic is $|E_1|/E_2 \le 2\%$. Two auxiliary versions are also used: a weak link ($U = 0$, dot replaced by a bond t_J) biased by $\Phi_0/2$, for which we take $L = 14$ because the single-electron harmonic of a homogeneous ring decays only as $e^{-L/\xi}$ ($|E_1/E_2| = 1\%$ at $L = 14$ against 8% at $L = 10$), and the short ring with a Zeeman-polarized dot of Ref. [1]. We write Eq. (1) compactly as $H = \hbar\omega a^\dagger a + H_{\rm ring}[\varphi_{\rm cl} + g(a + a^\dagger)]$.

The only property of $H_{\rm ring}$ that enters the adiabatic theory below is its ground-state energy at fixed phase and fixed fermion parity s,

$$E_s(\varphi) = \sum_{m\ge0} E_m \cos m\varphi. \tag{2}$$

Even harmonics ($m = 2, 4, \dots$) describe Cooper-pair tunnelling ($h/2e$ periodicity). The $m = 1$ term is the single-electron (h/e) persistent current of a mesoscopic ring [2] and is suppressed as $e^{-L/\xi}$. A 0-junction has $E_2 < 0$ and a π-junction has $E_2 > 0$. When a 0-junction is biased by $\Phi_0/2$, $\varphi_{\rm cl} = \pi/2$ maps $E_2\cos2\varphi \to -E_2\cos2\varphi$. The two routes to a π loop are therefore equivalent up to the odd harmonics, which turn into an explicit symmetry-breaking term $-E_1\sin\varphi$. In contrast, an intrinsic π-junction keeps $E(\varphi) = E(-\varphi)$ exactly, since $H_{\rm ring}[0]$ is real (time-reversal symmetry).

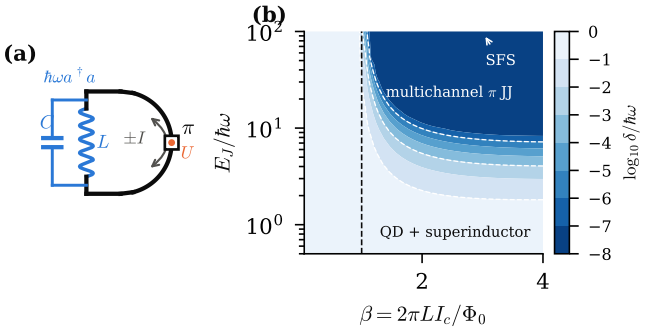


FIG. 1. (a) A superconducting loop closed by a π junction (here an interacting dot with charging energy U) shares its inductance L with an LC mode; the two degenerate states carry opposite persistent currents $\pm I$. (b) Tunnel splitting δ between the two current-carrying states in the plane of the screening parameter β and the quantum parameter $E_J/\hbar\omega$ [Eq. (5) with the DMRG $E(\varphi)$ of the $U = 2\Delta$ junction, $L = 10$]. White lines: $\delta/\hbar\omega = 10^{-6}, 10^{-3}, 10^{-1}$; dashed: classical threshold $\beta = 1$. Labels: typical regimes of the platforms discussed in the text.

Gauge-invariant current and sum rule. Since the Peierls phase of the loop is placed on the dot–ring bond, the persistent current is the derivative of H with respect to the total flux, $I = -\partial H/\partial\Phi = -(e/\hbar)\,\partial H/\partial\varphi_{\rm cl}$ (the Peierls phase is the single-electron phase, $\Phi = \hbar\varphi/e$),

$$I = \frac{e}{\hbar}\, i\, t_J \sum_{\sigma} \left(e^{i\varphi_{\rm cl}} e^{ig(a+a^\dagger)} c^\dagger_{1\sigma} c_{0\sigma} - \text{h.c.} \right) \tag{3}$$

(in terms of the Cooper-pair phase $\theta = 2\varphi$ the same operator reads $-(2e/\hbar)\partial H/\partial\theta$). This is the particle current through the dot–ring bond, dressed by the displacement operator of the mode: the combination $e^{i(\varphi_{\rm cl}+gX)} c^\dagger_1 c_0$ is invariant under the simultaneous gauge transformation of the electrons and of the vector potential, and its expectation value does not change if the Peierls phase is distributed along the ring with the pairing amplitude transformed covariantly [3]. Because the dot carries no pairing term, the particle number of site 0 is conserved by H and the current entering the dot from site $L-1$ equals the current (3) leaving it; the same is not true of the bonds inside the BCS chain, where the fixed-phase

pairing term acts as a condensate reservoir [3]. Equation (3) obeys an exact sum rule. With $X = a + a^\dagger$ and $P = -i(a - a^\dagger)$, $[H, P] = i\hbar\omega X - 2ig\,(\hbar/e)I$, so that in any eigenstate

$$\langle I \rangle = \frac{e}{\hbar}\,\frac{\hbar\omega}{2g}\,\langle a + a^\dagger \rangle : \tag{4}$$

the persistent current *is* the static displacement of the mode (the dc flux it generates, $\Phi = LI$). The two sides are a fermionic and a photonic observable, and only an eigenstate satisfies Eq. (4); we evaluate both independently and use their difference as the acceptance test of every DMRG state (Methods).

Adiabatic reduction and Landau theory. For $\hbar\omega$ well below the Andreev gap E_A, the electrons follow the flux adiabatically. To leading order in $\hbar\omega/E_A$, Eq. (1) reduces, in each parity sector, to a single-particle problem for the dimensionless flux $X = a + a^\dagger$,

$$H_{\rm BO} = \frac{\hbar\omega}{4}\left(P^2 + X^2\right) + E_s(\varphi_{\rm cl} + gX), \qquad [X, P] = 2i. \tag{5}$$

Equation (5) is the Hamiltonian of an rf-SQUID [4] whose inductive energy is supplied by the mode. Writing it in the Cooper-pair phase $\theta = 2gX$ gives $4E_C n^2 + \frac{1}{2}E_L\theta^2 + E(\theta/2)$ with $E_L = \hbar\omega/8g^2$ and $E_C = \hbar\omega g^2$. A static field ($\langle a \rangle \neq 0$), i.e. a photon condensate carrying a persistent current $I = -(e/\hbar)\partial_\varphi E$, appears classically when the orbital response of the ring at the symmetric point is paramagnetic and exceeds the stiffness of the mode,

$$\beta \equiv \frac{2g^2\,|E''(\varphi_0)|}{\hbar\omega} > 1, \qquad E''(\varphi_0) < 0. \tag{6}$$

For a pure π-junction, $\beta = 8g^2E_J/\hbar\omega = 2\pi LI_c/\Phi_0$ with $E_J \equiv E_2$, the familiar screening parameter of the loop [5, 6]. Equation (6) is the single-loop form of the photon-condensation criterion of Andolina *et al.* [7]: the no-go theorem for a uniform cavity field [8, 9] does not apply, because the mode couples to the flux (curl of $\mathbf{A}$), and the instability requires a *paramagnetic* orbital response, which here comes from the π shift. Above threshold, the Landau free energy $\mathcal{F}(X) = \hbar\omega X^2/4 + E(gX)$ has two degenerate minima $\pm X^\star$ carrying opposite currents $\pm I^\star$. Below threshold, in a 0 loop ($E''(0) > 0$), the same expansion gives a mode that is not softened but stiffened, $\omega \to \omega' = \omega\sqrt{1 + 2g^2E''(0)/\hbar\omega}$, whose ground state is a squeezed vacuum with $\langle a^\dagger a \rangle = (\omega'/\omega + \omega/\omega')/4 - 1/2$ virtual photons.

Two remarks fix the physical content of Eq. (6). First, for a loop with its own geometric inductance L_r and mutual inductance $M \leq \sqrt{L_rL_c}$ to the resonator, minimizing the static energy over the resonator flux gives $\Phi_r^2/2L_r$: the static threshold is set by the total loop inductance, and Eq. (1) is the $k = M/\sqrt{L_rL_c} \to 1$ (galvanic) limit. Second, zero-point fluctuations of the mode do not drive the transition; they oppose it. At finite $E_J/\hbar\omega$ the ground state of Eq. (5) is the symmetric superposition of the two current-carrying states, split from its partner by a tunnelling gap $\delta \sim \hbar\omega\, e^{-S}$ with an instanton action $S \propto (E_J/\hbar\omega)\, s(\beta)$. In the exact solution below this is quantitative: the threshold lies *above* the classical line $\beta = 1$ by $\Delta g = 0.02$, 0.04 and 0.12 as $E_J/\hbar\omega$ falls from 10 to 5 and 3.2, the crossover broadens from one grid step to $\Delta g \simeq 0.08$ at $E_J/\hbar\omega = 3.2$, and below $E_J/\hbar\omega \simeq 2.5$ no ordered phase is left for $g \leq 0.4$. A sharp quantum phase transition with mean-field exponents is recovered only in the scaling limit $E_J/\hbar\omega \to \infty$ at fixed β, in which $E_C/E_J = \beta(\hbar\omega)^2/(8E_J^2) \to 0$. This is the analogue of the classical-oscillator limit of the quantum Rabi model [10].

Methods. We solve Eq. (1) without approximation by two-site DMRG (TeNPy [11]) on the full electron–photon Hilbert space. The ring is stored in folded order next to a bosonic site (Fock cutoff $N_{\rm ph} = 50$–62, chosen such that the last Fock state carries a weight $< 10^{-7}$). Fermion parity and S^z are conserved, and the electron–photon vertex is a three-site operator $D\,c^\dagger_{1\sigma}c_{0\sigma}$. Benchmarks against exact BdG energies of the non-interacting ring at $g = 0$ agree to 10^{-14} at full bond dimension ($\chi = 1024$); for a 4-site ring with a photon, DMRG reproduces exact diagonalization and the sum rule (4) to 10^{-7}. In production $\chi = 300$ (truncation error $< 2 \times 10^{-8}$; $\chi = 400$ changes the energies by $< 4 \times 10^{-8}\Delta$ and the current by $< 0.3\%$ away from the crossover points). The accuracy of the energy is, however, not the relevant criterion. The two current-carrying states are split only by the exponentially small δ and by the seed, $2I^\star\varphi_{\rm cl} \sim 10^{-5}\Delta$, so that the energy landscape between the vacuum and the polarized state is flat on the scale of 10^{-7}–$10^{-6}\Delta$ that decides the polarization, while the local updates of DMRG drift towards the polarized state over tens of sweeps and can stop in an under-polarized state whose energy is within $10^{-6}\Delta$ of the ground state and whose current is several times too small. An earlier version of our scans suffered from this and, in addition, evaluated the observables after a warm start with the displacement operator of the previous g: the apparent poor convergence of the bond-current operator, which we had attributed to a cancellation of large terms, was this error, and the sum rule (4) exposes both. All DMRG results below therefore follow one protocol. (i) Every point is started from two independent states: a warm start from the converged neighbouring point on the vacuum side (A), and a product state with the mode in a coherent state at the classical flux, converged first with a larger seed ($\varphi_{\rm cl} = 10^{-2}$, then 3×10^{-3}) and then relaxed at the final seed (C), which approaches the ground state from the polarized side. (ii) The lower energy is kept whenever the two differ by more than $3\times10^{-7}\Delta$; otherwise the pair is reported as a bracket. (iii) A state is accepted only if it satisfies Eq. (4) to better than $10^{-3}I_c$ (median $10^{-4}I_c$ over all points with $g \geq 0.04$). (iv) The current above thresh-

old must be independent of the seed. Excited states are obtained by DMRG orthogonal to the ground state. We compare with (i) the self-consistent electron×photon product state (mean field, MF), the method of Ref. [1] generalized to the interacting dot (Hartree–Fock BdG electrons times a photon product state, swept upwards in g with the same seed), and (ii) the adiabatic Hamiltonian (5), with $E_s(\varphi)$ from BdG or from DMRG at fixed φ and $g = 0$, diagonalized exactly in a Fock space of 200 states.

Interaction-driven π junction. We first characterize the junction alone ($g = 0$). DMRG in the even and odd fermion-parity sectors [Fig. 2(b,c)] gives the 0–π transition known for magnetic quantum dots [12–18]. For small U the singlet (even) state is the ground state at every phase and the junction is a 0-junction; for large U the Kramers doublet (odd sector) is the ground state at every phase and $E(\varphi)$ has its *minimum* at the Cooper-pair phase π (electron phase $\varphi = \pi/2$). At $U = 2\Delta$ we find $E_2 = +7.39 \times 10^{-3}\Delta$ with $|E_1|/E_2 = 2\%$ and $|E_4|/E_2 < 1\%$: a pure π junction, indistinguishable after normalization from the half-flux-biased weak link [Fig. 2(a)]. The π coupling decreases monotonically with U ($E_2 \propto U^{-1.3}$ for $4\Delta \le U \le 8\Delta$), as expected for cotunnelling through the singly occupied level.

Three crossings of the two sectors matter below, and they are distinct [Fig. 2(b)]. The zero-flux crossing $U_c^0 = 0.94\Delta$ is the 0–π transition of the junction itself, where $E_{\rm odd}(0) = E_{\rm even}(0)$. The crossing of the *global* minima of the two surfaces, $E_{\rm odd}(\pi/2) = E_{\rm even}(0)$, lies lower, at 0.85Δ, because the doublet surface has its minimum at half a flux quantum while the singlet surface has it at zero. In between, $0.85\Delta < U < U_c^0$, the junction is a 0-junction at zero flux although the lowest state of the ring at *any* flux is the doublet: a loop in which the flux is free to adjust can switch to the doublet by letting a flux $\Phi_0/2$ in. This is what the mode does above a finite coupling, and it moves the boundary $U_c(g)$ of the loop into this window (next section). Finally, the singlet surface itself has a cusp at $\varphi = \pm\pi/2$ [dotted line in Fig. 2(a)]: at half a flux quantum two even-parity states cross so that $E_{\rm even}(\varphi)$ cannot be represented by a few harmonics of Eq. (2). The doublet surface, in contrast, is a smooth $\cos 2\varphi$ with 2% corrections.

Choice of regime and dimensionless variables. The microscopic model (1) contains the energy scales t, Δ, t_J, U of the junction and $\hbar\omega, g$ of the mode. Three dimensionless ratios control the physics. (i) U/Δ at fixed Γ/Δ sets the magnetic state of the dot and hence the sign of the Josephson coupling: the loop is a π loop only for $U > U_c$. (ii) The coupling g is fixed by the impedance of the mode alone, $g^2 = \pi Z/4R_Q$, i.e. $g = 0.1$–0.4 for $Z = 80\,\Omega$–$1.3\,{\rm k}\Omega$, from ordinary lumped resonators to the high-kinetic-inductance (superinducting) ones [19, 20]. (iii) $E_J/\hbar\omega$, with $E_J \equiv E_2 > 0$ the Josephson energy of the π junction, decides whether the transition is sharp: the two current states tunnel with $\delta \sim \hbar\omega e^{-S}$, $S \propto E_J/\hbar\omega$. We take $\hbar\omega = E_J(U = 2\Delta)/10 = 7.39 \times 10^{-4}\Delta$. In physical units this is the regime of a multichannel or gate-tunable π junction with $I_c \simeq 100\,{\rm nA}$ ($E_J/h \simeq 50\,{\rm GHz}$) in a loop that shares its inductance with a 5 GHz mode: $g = 0.1$–0.4 then corresponds to a loop inductance $L = Z/\omega = 3$–$40\,{\rm nH}$, and the screening condition $\beta = 2\pi L I_c/\Phi_0 = 1$ is met at $L = \Phi_0/2\pi I_c = 3.3\,{\rm nH}$, i.e. at $g = 0.11$ [class (iii) of the platforms discussed at the end]. With this choice the classical threshold $\beta = 8g^2 E_J(U)/\hbar\omega = 1$ lies at $g = 0.11$ for $U = 2\Delta$ and sweeps across the experimentally accessible window $g \le 0.4$ as U is varied from U_c to 8Δ, while $E_J/\hbar\omega$ decreases from 15 to 2, so that the same diagram contains the sharp transition and its quantum washing-out. We present it in the plane $(U/\Delta, g)$; the right axis of Fig. 3 converts g to Z/R_Q.

Grid. The grid is $g = 0.02$–0.40 in steps of 0.01 (0.02 above 0.3) and $U/\Delta = 0, 0.5, 0.75, 0.85, 1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8$, 427 ground states in all (about 20 CPU hours), each obtained with the two-start protocol of the Methods ($\chi = 300$, seed $\varphi_{\rm cl} = 10^{-3}$). For $U \le \Delta$ both parity sectors are computed and the lower one is taken, which locates the 0–π boundary of the *loop* without any assumption. The two starts differ by less than $3 \times 10^{-7}\Delta$ in energy at 47 points (11% of the grid, all inside the crossover windows discussed below), which are reported as brackets [Fig. 5(b)]. The sum rule (4) is satisfied to better than $10^{-3} I_c$ at every point with $g \ge 0.04$ (the worst points are the first ones of each scan at $g \le 0.03$, where the current itself is $< 10^{-2} I_c$) [Fig. 5(a)]. Increasing χ to 400 changes the current by $< 0.3\%$ except at the crossover point; raising $N_{\rm ph}$ to 80 changes the energy by $< 10^{-9}\Delta$; and reducing the seed to 10^{-4} leaves the current above threshold unchanged to 0.4% while the current below threshold scales down by the factor 0.09 [Fig. 5(c)]: the current is spontaneous.

Effective circuit model. When $\hbar\omega$ is small compared to the Andreev gap E_A of the junction ($\hbar\omega/E_A \simeq 10^{-3}$ here), the fermions can be integrated out adiabatically: the mode moves in the exact ground-state energy of the ring at fixed phase and fixed parity, $H_{\rm eff}^{(s)} = \hbar\omega a^\dagger a + E_s(\varphi_{\rm cl} + gX; U)$, $s = {\rm even}, {\rm odd}$ [Eq. (5)], an rf-SQUID Hamiltonian whose inductive energy is supplied by the mode; parity conservation forbids any mixing of the two surfaces, and the ground state is the lower of the two sector ground states. The matching of parameters is unambiguous: $E_s(\varphi; U)$ is computed by DMRG at $g = 0$. The doublet surface is represented by four harmonics (nine phases); the singlet surface, because of its cusp at $\varphi = \pi/2$ [Fig. 2(a)], is represented by a spline through 25 phases. $H_{\rm eff}$ is then diagonalized exactly in a Fock space of 200 states, with the same seed. Its current is $I_{\rm eff} = -(e/\hbar)\partial_\varphi E_s$, the classical current of the junction averaged over the flux distribution of the mode; the omit-

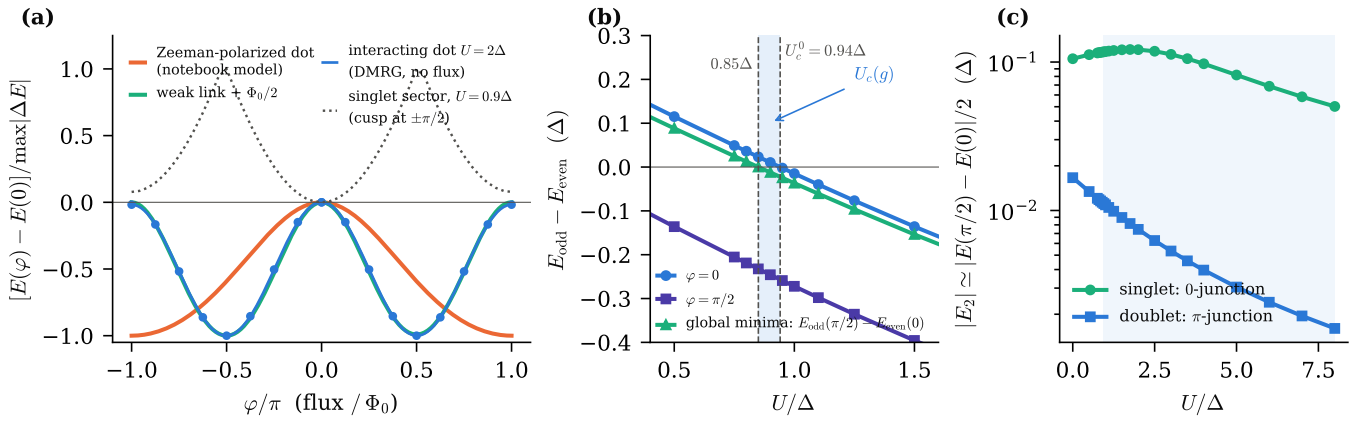


FIG. 2. (a) Normalized ground-state energy of the loop versus the electron phase φ (flux in units of $h/2e$) for three junctions: Zeeman-polarized dot in a short ring (dominated by the single-electron harmonic $\cos\varphi$), an ordinary weak link biased by $\Phi_0/2$ ($L=14$), and an interacting dot at $U=2\Delta$ without any bias ($L=10$; DMRG, symbols; line: cosine fit). Dotted: the singlet (even-sector) surface at $U=0.9\Delta$, with its cusp at $\pm\pi/2$. (b) Parity competition $E_{\rm odd}-E_{\rm even}$ of the interacting-dot ring at $\varphi=0$, at $\varphi=\pi/2$, and between the global minima of the two surfaces (DMRG, $L=10$, $\Delta=t$, $t_J=0.5\Delta$). The zero-flux crossing $U_c^0=0.94\Delta$ and the crossing of the global minima at 0.85Δ bound the window (shaded) in which the boundary $U_c(g)$ of the loop moves with the coupling to the mode. (c) Josephson energy $|E_2|$ in the singlet (0) and doublet (π) sectors. Shading: doublet ground state at zero flux.

ted terms are the non-adiabatic (Berry) corrections of relative order $\hbar\omega/E_A$. The mean-field theory of Ref. [1], generalized to the interacting dot (Hartree–Fock BdG electrons × photon product state, swept upwards in g with the same seed), is evaluated on the same grid.

Phase diagram. Figure 3 shows $|\langle I\rangle|/I_c(U)$, with $I_c(U)$ the critical current of the π junction at $g=0$, from the full DMRG (a), the effective circuit model (b) and mean field (c); Fig. 4 shows cuts. The exact diagram has three regions. (i) For $U<U_c(g)$ the loop is a 0 loop: the singlet is the ground state at all g, the orbital response is diamagnetic and the mode is not softened but stiffened, $\omega\to\omega'=\omega\sqrt{1+2g^2E''(0)/\hbar\omega}$ with $E''(0)=0.20\Delta$, i.e. $\omega'=9.4\,\omega$ at $g=0.4$. The ground state is the squeezed vacuum of the stiffened mode: its photon number $(\omega'/\omega+\omega/\omega')/4-1/2$ reproduces the DMRG value at every g (1.89 versus 1.91 virtual photons at $g=0.4$), while $|\langle I\rangle|<10^{-2}I_c$ and linear in the seed. The boundary is not the zero-flux 0–π crossing of the junction, $U_c^0=0.94\Delta$ [Fig. 2(b)]: the doublet gains the condensation energy of the π loop (up to $2E_J$), so that $U_c(g)$ decreases from 0.94Δ at $g\to0$ towards the crossing of the global minima, 0.85Δ, which it reaches at $g\simeq0.4$ [at $U=0.85\Delta$, $E_{\rm odd}-E_{\rm even}$ falls from $2.3\times10^{-2}\Delta$ at $g=0.02$ to $9\times10^{-4}\Delta$ at $g=0.4$]. The flux of the mode thus drives a parity switch of the junction, and in the window $0.85\Delta<U<0.94\Delta$, where the global minimum of $E(\varphi)$ already belongs to the doublet, the switch occurs at a finite coupling ($g\simeq0.3$ at $U=0.9\Delta$ in the effective model). This shift of U_c by 0.09Δ is a prediction specific to the microscopic treatment: the effective circuit model reproduces it only once the singlet surface, cusp included, is represented exactly. (ii) For $U>U_c$ and $g>g_c(U)$ the mode condenses: at $U=2\Delta$ the photon number rises from 0.5 at $g=0.11$ to 5.7 at $g=0.13$ and 7.3 at $g=0.14$, where the current reaches $0.96\,I_c$; it then decreases as $I\simeq(e/\hbar)\hbar\omega\varphi^\star/2g^2$, because the condensate flux saturates at $\Phi_0/2$ while the inductance $\propto g^2/\hbar\omega$ grows [Eq. (4)]. The threshold $g_c(U)$ (first grid point with $|\langle I\rangle|>0.5I_c$) rises from 0.10 at $U=\Delta$ to 0.13, 0.19, 0.24 and 0.32 at $U=2,4,5,6\Delta$, following the decrease of $E_J(U)$. It lies systematically *above* the classical line $\beta=1$ (dash-dotted), by $\Delta g=0.02$ at $U=2\Delta$, 0.04 at 4Δ and 0.12 at 6Δ: the zero-point motion of the flux opposes the condensation, and the shift grows as $E_J/\hbar\omega$ decreases from 10 to 3.2. At the same time the transition broadens: the width of the crossover in g grows from one grid step at $U\le3\Delta$ to 0.05 at $U=5\Delta$ and 0.08 at $U=6\Delta$, and the maximal current drops from $0.97I_c$ ($U=\Delta$) to 0.90, 0.81 and $0.61I_c$ ($U=4,5,6\Delta$). For $U=8\Delta$ ($E_J/\hbar\omega=2.1$) no ordered phase is left in the window $g\le0.4$ although β reaches 2.7: there the ground state is the cat state with $\delta\simeq\hbar\omega$ and $\langle I\rangle\le0.04I_c$, proportional to the seed. This is the finite-$E_J/\hbar\omega$ remnant of the transition, which becomes sharp only in the limit $E_J/\hbar\omega\to\infty$ at fixed β (Fig. 1).

Comparison with the effective model and with mean field. Above threshold the full DMRG and the effective circuit model agree to 0.2% of I_c on average and to 2.4% at worst over the whole ordered region, and to 0.1 in the photon number (the doublet-sector energies agree to better than $10^{-5}\Delta$ at all g, to $10^{-6}\Delta$ for $g\le0.3$), which validates the adiabatic elimination of the fermions at $\hbar\omega/E_A\sim10^{-3}$. The only points where they differ by

more are the bracket points inside the crossover windows, where the exact ground state is a partially polarized cat and the DMRG from the two initial states brackets the effective-model value from both sides [vertical bars in Fig. 4(a)]. This is a statement about the convergence of a variational method on a flat energy landscape, not about the physics: the energy difference between the two starts is $< 3\times10^{-7}\Delta$, while the effective model, which resolves it exactly, interpolates smoothly between them. Mean field locates $g_c(U)$ to within the grid step for $U \le 5\Delta$ (0.27 versus 0.32 at 6Δ) but underestimates the current by 10–20% (0.86 versus $0.96\,I_c$ at $U = 2\Delta$; 0.72 versus $0.90\,I_c$ at 4Δ) because the product state renormalizes the junction bond by $|\langle D\rangle| < 1$; it overestimates the photon number at large g (5.2 versus 3.7 at $U = 2\Delta$, $g = 0.4$), where the exact flux distribution is a displaced squeezed state rather than a coherent state; it predicts an ordered phase with $0.6I_c$ at $U = 7$–8Δ that the exact solution does not have, because a product state cannot describe the tunnelling between the two current directions that restores the symmetry when $E_J \lesssim 3\hbar\omega$; and, since its Hartree–Fock threshold $U_c^{\mathrm{MF}} = 0.82\Delta$ replaces the Kramers doublet by a static moment, it orders the loop at $U = 0.85\Delta$ with $0.95I_c$ where the exact ground state is the singlet 0 loop. In short, mean field gives the right g_c and a 10–20% low current where $E_J \gg \hbar\omega$ and U is well above U_c, and fails qualitatively at both ends of the diagram, $U \gtrsim 7\Delta$ and $U \simeq U_c$.

Cat state and macroscopic tunnelling of the π loop. Without a seed ($\varphi_{\mathrm{cl}} = 0$ exactly) H is real, $\langle I\rangle = 0$ in every eigenstate, and the order is diagnosed by the two lowest states of the doublet sector [Fig. 6, $U = 2\Delta$, $E_J/\hbar\omega = 10$, the parameters of the $U = 2\Delta$ cut of Fig. 4]. The splitting $\delta = E_1 - E_0$ collapses and the flux operator projected onto the quasi-degenerate doublet acquires the eigenvalues $\pm\Phi^\star$. DMRG and the adiabatic theory give the same $\Phi^\star$ to within 0.2% at all β [$\Phi^\star = 0.59\,\Phi_0/2$ at $\beta = 2$], the same photon number to 0.03, and the same $\delta/\hbar\omega = 0.70$, 0.24 and 2.65×10^{-4} at $\beta = 0.5$, 1 and 1.5; at $\beta = 1.5$ the DMRG values at $\chi = 300$ and 400 are 2.66 and 2.65×10^{-4}, so that the macroscopic tunnelling of a condensate of seven photons ($\delta = 2\times10^{-7}\Delta$) is resolved directly. Deeper in the ordered phase δ drops below the DMRG resolution for excited-state energies ($4\times10^{-5}\,\hbar\omega$, the change of the energies between $\chi = 300$ and 400); the adiabatic theory, validated above, then gives $\delta/\hbar\omega = 8\times10^{-7}$ and 7×10^{-9} at $\beta = 2$ and 3. The exact ground-state energy lies $(0.4$–$1.2)\times10^{-3}\hbar\omega$ above the adiabatic one, the non-adiabatic correction of relative order $\hbar\omega/E_A$. The seed-free DMRG ground state is the cat state itself: $|\langle\Phi\rangle| < 10^{-3}\Phi^\star$, the flux distribution $P(\Phi)$ is bimodal with peaks at $\pm\Phi^\star$ whose separation tracks the classical minimum [Fig. 6(c)], and a DMRG started from the polarized coherent state at $+\Phi^\star$ relaxes to the same symmetric state (energy difference $< 10^{-10}\Delta$). This is the opposite situation to the seeded scans of Fig. 3, where the seed selects one well and the two-start protocol is needed to reach it. With increasing $E_J/\hbar\omega$ the crossover sharpens into a pitchfork at $\beta = 1$ and δ vanishes exponentially [Fig. 6(b)]. The residual δ is the macroscopic tunnelling of a persistent current. Because it is protected by time-reversal symmetry, it is the only energy scale separating the two current directions. Each current state is additionally Kramers degenerate in the spin of the dot. The low-energy manifold is therefore a fourfold multiplet $\{\sigma =\uparrow,\downarrow\}\otimes\{\circlearrowleft,\circlearrowright\}$, split only by δ.

Half-flux-biased weak link. For an ordinary weak link ($L = 14$, $t_J = 0.5$, $E_2 = -0.056\,\Delta$, $E_1 = -5.4\times10^{-4}\Delta$, Andreev gap $0.77\,\Delta$) biased by $\Phi_0/2$, Fig. 7 shows the full DMRG solution at $E_J/\hbar\omega = 10$ as a function of β, recomputed with the two-start protocol (the sum rule (4) holds to $2\times10^{-4}I_c$ at every point and to $6\times10^{-5}I_c$ above threshold). The current rises steeply at $\beta \simeq 1.1$ and saturates at $|\langle I\rangle| \simeq 0.96I_c$ for $\beta \approx 1.7$; the recomputed currents differ from the earlier warm-started runs by at most $0.02I_c$ (at $\beta = 1.1$–1.4), so this result was not affected by the under-polarization that spoiled the seeded scans at the symmetric point. It then decreases as the flux approaches $\Phi_0/2$, where the junction current vanishes. DMRG and the adiabatic theory agree to $2\times10^{-3}\hbar\omega$ in energy, to 2% of I_c in current and to 1.3% in $E_1 - E_0$ at all β; the vacuum-side and polarized-side starts give the same flux to 0.3% and the same energy to $10^{-6}\Delta$, because the tilt of the double well by the odd harmonic makes the landscape better conditioned than at the symmetric point. The product state is 0.2–$0.4\hbar\omega$ too high in energy and overestimates the current threefold near threshold ($0.71I_c$ versus $0.22I_c$ at $\beta = 1.13$). It becomes accurate only deep in the ordered phase, where the photon is a nearly coherent state. The flux distribution extracted from the photon reduced density matrix evolves from a Gaussian centred at zero to a single displaced peak at $\pm\Phi^\star$. The sign is fixed by the residual odd harmonic E_1, which tilts the double well by $2|E_1|\sin\varphi^\star \approx 10^{-3}\Delta \approx 0.2\hbar\omega$. This tilt sets the splitting $E_1 - E_0$ for $\beta > 1.2$ in DMRG and BO alike. Without it, $\delta/\hbar\omega$ would drop below 10^{-6} at $\beta = 2$. The external-bias route thus provides a current-carrying state whose direction is *selected* rather than spontaneously chosen. Its residual symmetry breaking is $\propto e^{-L/\xi}$ for a homogeneous ring, but in practice it is limited by the flux offset $\delta\Phi$.

Breakdown of the product-state description for $E_J \simeq \hbar\omega$. The product state is reliable only when $E_J \gg \hbar\omega$ and $g \ll 1$, and both conditions are violated by the parameters of the original mean-field solution [1]: a Zeeman-polarized dot embedded in a short BCS ring ($t = 2$, $\Delta = 0.6$, $L = 11$, $\epsilon_{d\uparrow} = -0.3$, $\epsilon_{d\downarrow} = 3.7$, $t_J = 0.5$, $\hbar\omega = 0.02$; energies in units of t in this paragraph). The self-consistent product state predicts a current-carrying phase for $g \gtrsim 0.75$, with $\langle a^\dagger a\rangle \simeq 1$–$1.4$ and $|\langle I\rangle| \approx 0.02$ [Fig. 8]. DMRG shows that this phase is an artefact.

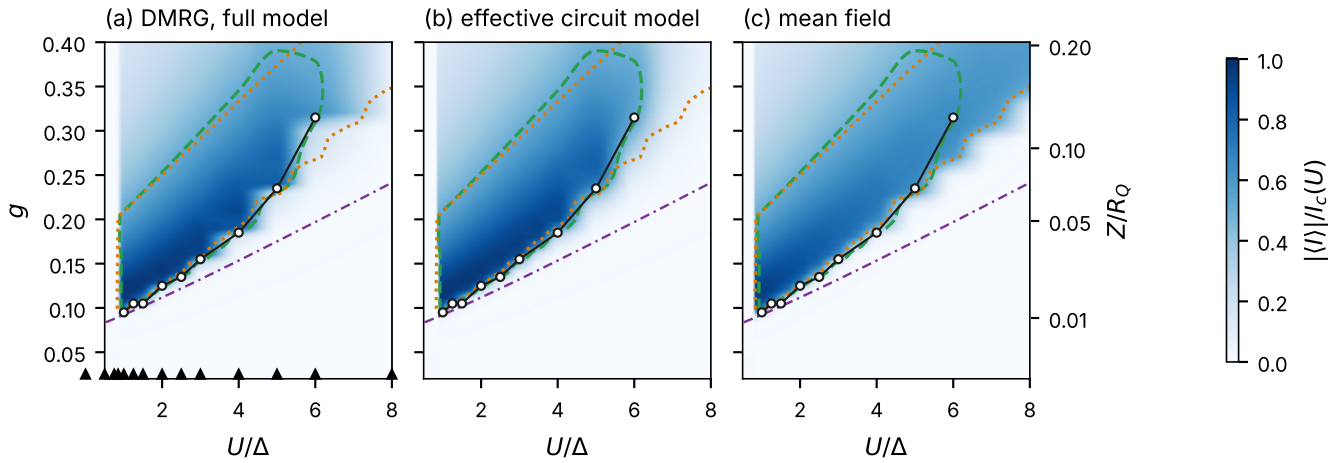


FIG. 3. Phase diagram of the microscopic model (1) in the plane of the dot charging energy U/Δ and the mode coupling g ($Z/R_Q = 4g^2/\pi$, right axis), at $\hbar\omega = 7.39\times10^{-4}\Delta$, $t_J = 0.5\Delta$, $L = 10$, no external flux (seed $\varphi_{\rm cl} = 10^{-3}$). Colour: persistent current $|\langle I\rangle|$ from the gauge-invariant operator (3) in units of the critical current $I_c(U)$ of the π junction. (a) Full DMRG of electrons and photon, both parity sectors (triangles: computed U values; g step 0.01). (b) Effective circuit model $H_{\rm eff}$ with the DMRG $E_s(\varphi;U)$. (c) Mean field. Circles: DMRG threshold ($|\langle I\rangle| = 0.5I_c$, bar: grid step); green dashed and orange dotted: the same contour from the effective model and from mean field; purple dash-dotted: classical threshold $\beta = 1$.

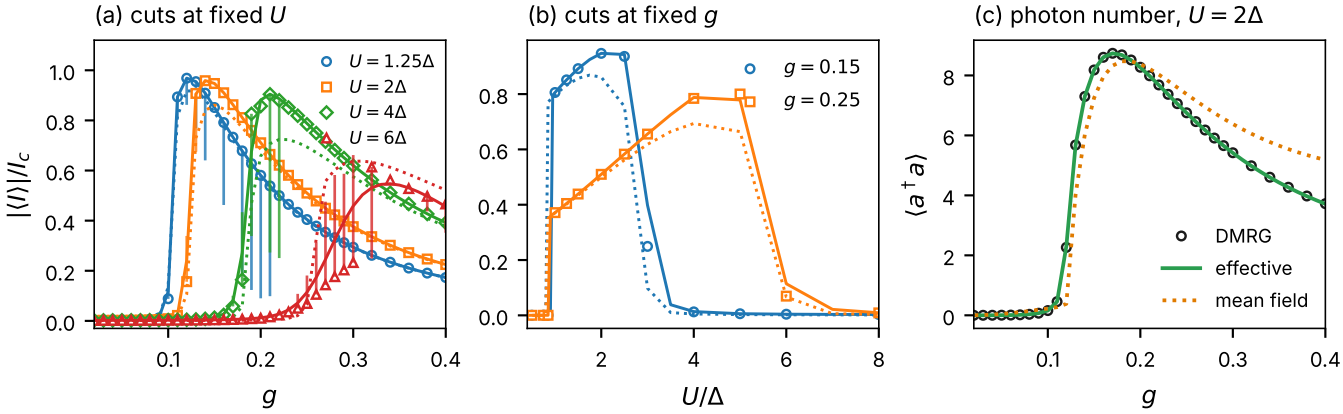


FIG. 4. Cuts through Fig. 3. (a) Current versus g at $U/\Delta = 1.25, 2, 4, 6$ and (b) versus U at $g = 0.15, 0.25$: full DMRG (symbols; vertical bars: bracket between the two initial states, converted to a current with Eq. (4)), effective circuit model (solid), mean field (dotted). (c) Photon number at $U = 2\Delta$.

The exact ground-state energy decreases monotonically with g and lies below the MF energy by up to $1.8\hbar\omega$ (at $g = 1.2$). The gap to the first excited state stays a sizeable fraction of $\hbar\omega$ ($E_1 - E_0 = 0.33\hbar\omega$ at $g = 1.2$), and the photon number saturates at $\langle a^\dagger a\rangle \simeq 0.36$. The response to the seed flux $\varphi_{\rm cl} = 10^{-3}$ is linear: $\langle I\rangle \approx 7\times10^{-5}$, more than two orders of magnitude below the MF value. DMRG and the adiabatic theory (5) agree to within $0.06\hbar\omega$ in energy and 1.3% in $E_1 - E_0$. The residual difference is the non-adiabatic (diagonal Born–Oppenheimer) correction, of relative order $\hbar\omega/E_A \approx 0.1$. At $\hbar\omega = 0.002$ the same microscopic model does order, at $g_c \simeq 0.21$, while mean field predicts $g_c \simeq 0.30$ [Fig. 9].

Two effects make the product state fail. (i) It is not gauge invariant [3, 21]: with the full Peierls phase on one bond, it renormalizes that bond by $|\langle D\rangle| < 1$ and so loses phase-*independent* kinetic energy, which the exact state recovers by following the flux adiabatically. (ii) Here $E_J \simeq \hbar\omega$. The Landau free energy does develop a double well for $g > 0.58$, but its depth is below the zero-point energy, and the flux distribution $P(X)$ stays essentially unimodal. A further subtlety concerns ring size: since $L \lesssim \xi$, $E(\varphi)$ of this ring is dominated by the single-electron harmonic, $E_1 = 2.1\times10^{-2} \gg E_2 = 2.0\times10^{-3}$ [Fig. 2(a)]. The instability would then be one of the normal h/e persistent current rather than of the Josephson junction, which is why all other results use rings with $L \gg \xi$.

Experimental scales and signatures. The two control parameters are $\beta = 2\pi LI_c/\Phi_0$ and $E_J/\hbar\omega$, related by

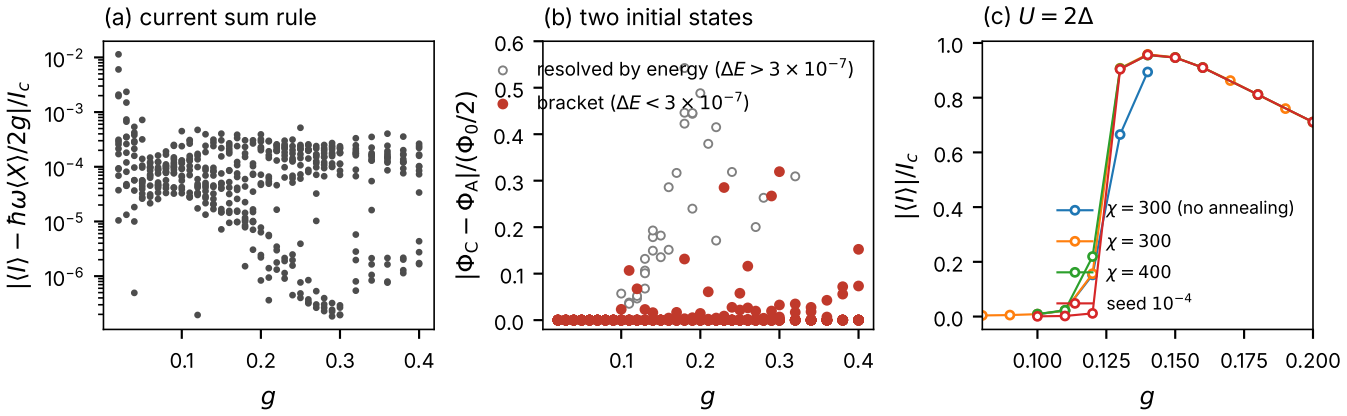


FIG. 5. Convergence of the full DMRG. (a) Violation of the sum rule (4) over the whole grid. (b) Difference between the fluxes of the two initial states (vacuum side A and polarized side C) in units of $\Phi_0/2$: open symbols, resolved by the energy; filled, reported as a bracket. (c) $U = 2\Delta$ near threshold: $\chi = 300$ without and with the annealed start, $\chi = 400$, and seed 10^{-4}.

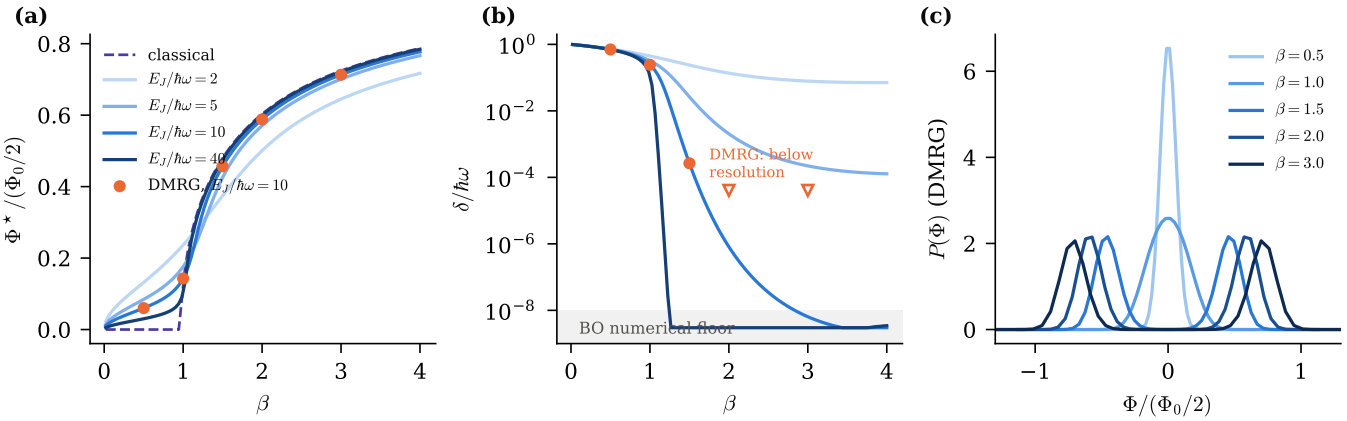


FIG. 6. Interaction-driven π loop ($U = 2\Delta$, $L = 10$, $\hbar\omega = E_J/10$, no external flux, no seed). (a) Flux of the condensate $\Phi^\star$ (eigenvalue of the flux projected on the two lowest states) versus β for increasing $E_J/\hbar\omega$ (lines: adiabatic theory with the DMRG $E(\varphi)$; circles: full DMRG at $E_J/\hbar\omega = 10$; dashed: classical). (b) Tunnel splitting δ between the two current-carrying states (triangles: DMRG below its resolution). (c) Flux distribution of the DMRG ground state (a cat state for $\beta > 1$).

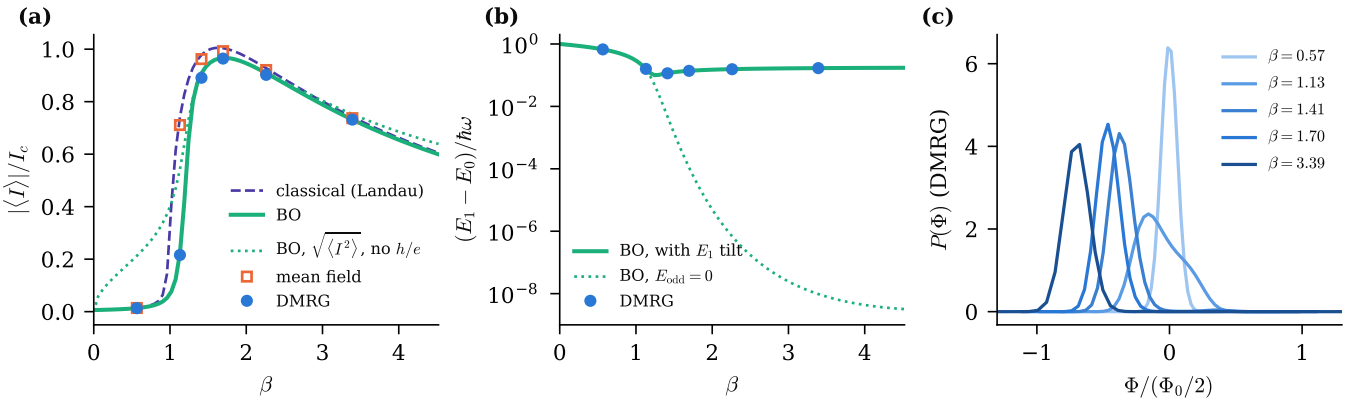


FIG. 7. Weak-link loop ($L = 14$) biased by $\Phi_0/2$, $E_J/\hbar\omega = 10$. (a) Persistent current, Eq. (3), versus β: DMRG with the two-start protocol (circles), MF (squares), adiabatic theory with (solid) and without (dotted) the odd harmonics, classical Landau theory (dashed). (b) Splitting of the two lowest levels. (c) Flux distribution of the DMRG ground state, obtained from the photon reduced density matrix.

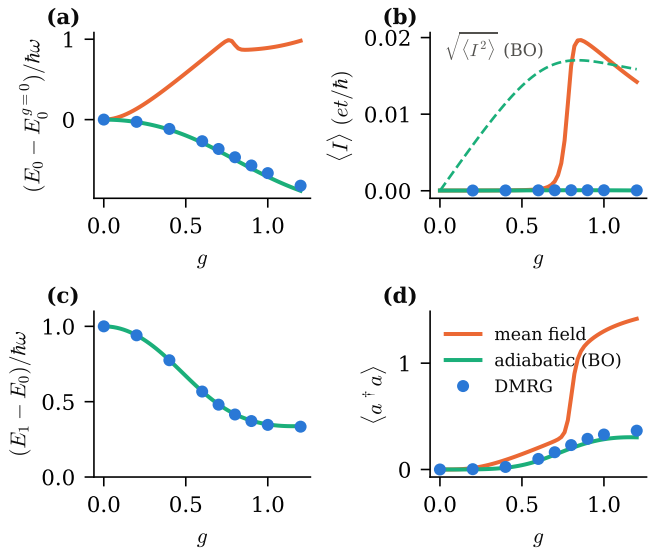


FIG. 8. Short ring with a Zeeman-polarized dot ($\hbar\omega = 0.02$, $\varphi_{\rm cl} = 10^{-3}$): mean field (MF, orange), adiabatic theory (5) (green), and DMRG (symbols). (a) Ground-state energy, (b) current (dashed: rms current of the BO ground state), (c) gap to the first excited state, (d) photon number.

$\beta = 2\pi(Z/R_Q)(E_J/\hbar\omega)$, with $E_J/h \simeq 0.5\,\mathrm{GHz} \times I_c[\mathrm{nA}]$. Three classes of π elements span the phase diagram [Fig. 1(b)]: (i) SFS junctions ($I_c \sim$ 1–100 μA) reach $\beta > 1$ with geometric inductances of a few to a few hundred pH and $E_J/\hbar\omega \sim 10^2$–$10^4$. This is the classical limit, where the half-flux state is fully polarized, consistent with the spontaneous currents observed in SFS and d-wave π loops [6, 22, 23]. (ii) Quantum-dot π junctions in the doublet regime ($I_c^\pi \sim 0.1$–$10\,\mathrm{nA}$) require $L \sim 0.03$–$3\,\mu$H, i.e. superinductances (Josephson-junction arrays, granular Al) as in fluxonium circuits [19, 20]; this is the π-qubit regime of Refs. [24–26]. These give $E_J/\hbar\omega \sim 1$–3, and the exact diagram is unambiguous there: for $E_J/\hbar\omega \lesssim 2.5$ no current-carrying phase exists at any coupling $g \le 0.4$, even where β reaches 2.7 (the $U = 8\Delta$ line of Fig. 3). The ground state is the cat state $(|\circlearrowleft\rangle + |\circlearrowright\rangle)/\sqrt{2}$ with $\delta \sim \hbar\omega$, and the crossover appears as a dip in the $|0\rangle \to |1\rangle$ frequency and a jump in the transition matrix element $\langle 0|\Phi|1\rangle$, accessible in two-tone spectroscopy. A mean-field description predicts a persistent current of $0.6I_c$ in this regime; it is absent. (iii) Multichannel or gate-tunable π junctions with $I_c \sim$ 50–200 nA in a high-kinetic-inductance loop ($L \sim$ 3–40 nH, $\omega/2\pi \sim$ 2–5 GHz) give $\beta \sim$ 1–10 and $E_J/\hbar\omega \sim$ 10–30. This is the regime of the phase diagram (Figs. 3, 4) and of Figs. 6 and 7, worked out above for $I_c = 100\,\mathrm{nA}$ and a 5 GHz mode: a well-defined current-carrying state of $\simeq 0.9I_c$ at $\beta \simeq$ 1.3–2, whose residual tunnelling $\delta/\hbar\omega \sim 10^{-4}$–$10^{-8}$ remains a measurable macroscopic-quantum-tunnelling signature, and a threshold that lies 10–20% above the classical $\beta = 1$ in g. In the externally biased realization, the half-flux bias must be stable to $|\delta\Phi| \ll \delta/2I^\star$, i.e. $\lesssim 10^{-6}\Phi_0$ for the parameters above. The interaction-driven realization has no such requirement.

Two signatures distinguish the microscopic π loop from a classical π loop with a fixed Josephson energy. First, the 0 phase is not inert: for $U < U_c(g)$ the mode is stiffened, $\omega'/\omega = \sqrt{1 + 2g^2E''(0)/\hbar\omega}$ up to 9.4 at $g = 0.4$ here, and its ground state is a squeezed vacuum whose photon number reproduces the DMRG value to 1%. A resonator frequency that *rises* with the coupling in the 0 phase and softens towards the condensation in the π phase tells which side of U_c the junction is on, independently of the current, and is directly accessible in transmission. Second, the mode itself moves the 0–π boundary: a gate that tunes U/Γ across the window $0.85\Delta < U < 0.94\Delta$ switches the parity of the junction at a coupling-dependent point $U_c(g)$, 0.09Δ below the zero-flux crossing at $g = 0.4$. This parity switch is a prediction of the microscopic treatment; the effective circuit model reproduces it only with the exact singlet surface, and mean field, which replaces the Kramers doublet by a static moment, places it at 0.82Δ for every g.

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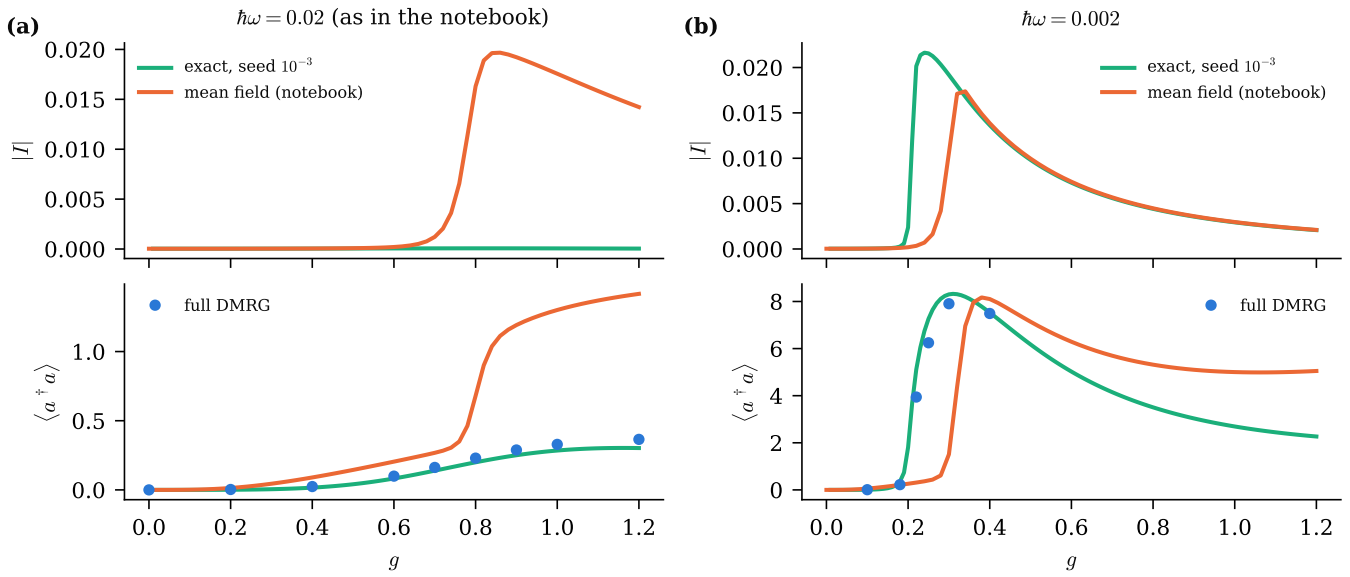


FIG. 9. Model of the original notebook (Zeeman-polarized dot, $L = 11$, $t = 2$, $\Delta = 0.6$): current (top) and photon number (bottom) versus g for (a) $\hbar\omega = 0.02$ and (b) $\hbar\omega = 0.002$. Green: exact (adiabatic, seed 10^{-3}); orange: mean field; circles: full DMRG (photon number; these older runs predate the two-start protocol).

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